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$$f(x, y) = \sqrt{50 - x^2 - y^2} \text{ in het punt } \mathbf{a}(4, -3)$$

$$f(a) = \sqrt{50 - 16 - 9} = \sqrt{25} = 5$$

$$\frac{\partial f}{\partial x} = \frac{-x}{\sqrt{50 - x^2 - y^2}}$$

$$\frac{\partial f}{\partial y} = \frac{-y}{\sqrt{50 - x^2 - y^2}}$$

$$\frac{\partial f}{\partial x}(a) = \frac{-4}{5}$$

$$\frac{\partial f}{\partial y}(a) = \frac{3}{5}$$

$$z = 5 - \frac{4}{5}(x - 4) + \frac{3}{5}(y + 3) = -\frac{4}{5}x + \frac{3}{5}y + 10$$

References